

**NAVSEA
OCCUPATIONAL SAFETY,
HEALTH AND ENVIRONMENT
CONTROL MANUAL
FOR NAVAL SHIPYARDS**

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CHAPTER 220 FALL PROTECTION

220.1 DISCUSSION:

A. Purpose: To provide a compilation of requirements applicable to the Corporate Fall Protection Program.

B. Standardization: This document provides the standardized Fall Protection Safety Policy for Naval Shipyards.

C. Scope: This document applies to evolutions, operations, or tasks that may expose personnel, either shipboard or ashore, who are accessing heights above a solid surface and exposed to fall hazards.

D. Applicability: This program applies to all personnel working at heights and/or utilizing fall protection equipment provided by Naval Shipyards and Intermediate Maintenance Facilities who are accessing heights where there is a possibility of falling:

1. Four feet for general industry work,
2. Five feet for ship repair operations, or
3. From any height onto dangerous equipment, into a hazardous environment, or onto an impalement hazard.

E. Background:

1. Falls from heights are a leading cause of work-related injuries and fatalities. The Naval Shipyards continue to experience serious fall-related mishaps, which lead to reduced readiness and productivity, as well as high medical and compensation costs resulting from these mishaps.

2. Mishaps involving falls are generally complex events, frequently involving a variety of factors. Consequently, preventing falls involves worker engagement, work procedures and fall protection system(s) design.

3. Every Navy command has a duty to protect its workforce by anticipating the need to work at heights and planning the work to protect from all hazards. Effective mishap prevention must be incorporated into the job planning process. All falls are preventable. Careful planning, preparation, and worker participation lay necessary groundwork for an accident-free workplace.

F. Effective Implementation Date.

The effective implementation date is 1 year from the issuance of this chapter.

220.2 APPENDICES:

A. Definitions.

B. Excerpt from OSHA Directive Number CPL 02-00-162 (Shipyard Employment “Tool Bag” Directive).

C. Example Engineered Anchorage Label.

D. Example Inspection Attributes Record.

E. Fall Hazard Survey and Protection Plan

220.3 REFERENCES:

A. OPNAV M-5100.23, Navy Safety and Occupational Health Program Manual, (Latest Revision).

B. OSHA Directive Number CPL 02-00-162 (Shipyard Employment “Tool Bag” Directive).

C. NAVFAC P-307, Management of Weight Handling Equipment.

D. 29 Code of Federal Regulations (CFR) 1910, Occupational and Health Standards for General Industry.

E. American National Standards Institute (ANSI)/American Society of Safety Professionals (ASSP) Z359 Fall Protection Code.

F. NAVSEAINST 4700.17, Preparation and Submission of Trouble Reports.

G. 29 Code of Federal Regulations (CFR) 1915, Occupational and Health Standards for Shipyard Employment.

H. Naval Ships Technical Manual (NSTM) Chapter 600, Hull Outfitting Equipment (Latest Revision).

I. Department of the Navy Fall Protection Guide (Latest Revision).

J. OPNAVINST 3500.39D Operational Risk Management (Latest Revision).

K. OSHE Control Manual 130, OSHE Deficiency Reporting System.

220.4 OVERVIEW OF SUBJECT MATTER AND SPECIFIC POLICY:

A. Definitions: To implement the requirements of this chapter effectively, numerous terms and definitions must be understood. For convenience, they are listed in Appendix A, Definitions.

B. General Requirements:

1. Due to the nature of work performed in Naval Shipyards and Intermediate Maintenance Facilities, workers at times will be required to perform their duties at elevated locations. Personnel exposed to fall hazards must be appropriately trained in accordance with this chapter. Fall hazards must be mitigated in accordance with this chapter.

2. Workers weighing less than 130 and over 310 pounds, as well as those with known back problems or heart conditions, are at a greater risk of personal injury from fall protection equipment in the event fall arrest occurs. These individuals must self-identify to their supervisor and the Fall Protection Program Manager prior to entering into the Fall Protection Program. More information on this topic is discussed within section 220.4.E.1 of this manual.

a. Currently, there is no fall protection equipment approved for use by pregnant workers. Pregnant workers must not don personal fall protection equipment.

3. Personnel must report, through their chain of command, when fall protection equipment is impacted. Personnel must be medically evaluated in the event fall arrest occurs.

4. The Hierarchy of Fall Protection per reference (E), or preferred order of control measures, for an identified fall hazard is:

a. Hazard Elimination: Changing the task, process, controls or other means to remove the need for an employee to be exposed to a fall hazard (e.g., bring the job to the ground).

b. Passive Fall Protection System: Use of a physical barrier that isolates the worker from the fall hazard, which does not require the wearing or use of personal fall protection equipment (e.g., guardrail systems, covers, safety nets, etc.).

c. Fall Restraint System: Restrict the worker's movements so the worker's center of mass cannot reach beyond the fall hazard by utilizing a combination of anchorage, anchorage connectors, connecting means and full body harness.

d. Fall Arrest System: The collection of equipment components that are configured to arrest a free fall by utilizing a combination of anchorage, anchorage connectors, connecting means (energy absorbing lanyard, SRD, etc.) and full body harness.

e. Administrative Controls: Within the fall protection plan; detail the infeasibility of the above listed fall protection mitigation measures, introduce new work practices that reduce the risk of falling, and outline how to perform work tasks as safely as possible without a secondary system (e.g., erecting warning lines or designated areas, restricting access to certain areas, or posting warning signs).

NOTE

Control measures are not mutually exclusive. There may be situations when more than one control measure should be used to reduce the risk of a fall.

5. Where fall hazards exist, workers must maintain a primary and secondary fall protection system at all times. If the primary fall protection system fails, the secondary fall protection system will protect the worker.
6. Personnel must not work at heights, work near unguarded edges, or work in an isolated area alone. Whenever possible, employees should work in pairs (buddy system). This is to ensure the Rescue Plan can be carried out as quickly as possible.
7. All unguarded edges must be evaluated and hazards mitigated in accordance with this chapter.
8. Fall protection requirements must be considered and used regardless of the time duration of the work. For example; if a 2-minute job requires 15 minutes to establish fall protection, then personnel must spend the required 15 minutes to establish their fall protection.
9. Fall Protection equipment must be turned into the tool room;
 - a. For inspection per section 220.4.G.2. (Inspection),
 - b. Once it has been impacted or shows signs of impact loading, or
 - c. When it fails pre-use/post-use inspection.

C. Specific Requirements:

1. Walking/Working Surfaces.
 - a. When personnel will be on a surface with unprotected sides or edges 4 feet (5 feet for ship repair operations) or greater above a lower level, they must be protected with Fall Protection Systems in accordance with this chapter.
 - b. Personnel exposed to fall hazards on walking-working surfaces at **any** height above dangerous equipment, impalement hazards, or hazardous environments must be protected with Fall Protection Systems in accordance with this chapter.
 - c. When any portion of a guardrail system, gate, or safety chain is removed and personnel are exposed to an unguarded edge, personnel must be protected from falling with Fall Protection Systems in accordance with this chapter.
 - d. When guardrail systems are used around holes that serve as points of access (e.g., ladderways, hatches, etc.), the guardrail system opening:

(1) Must possess a self-closing gate that slides or swings away from the hole, and meets the requirements of section 220.4.D.1 (Guardrail Systems), or

(2) Must offset to prevent personnel from walking or falling into the hole.

e. Reference B provides to Occupational Safety and Health Administration (OSHA) employees and interested parties' guidance concerning the application of occupational safety and health standards in shipyard employment. Appendix B, Excerpt from OSHA Directive Number CPL 02-00-162 (Shipyard Employment "Tool Bag" Directive), provides guidance concerning the applicability of 29 CFR 1910.28, 1910.29, and 1910.140 for fall protection on vessels and on shore.

2. Edges over/near water. When working at the unguarded edge of a pier, quay wall, or flooded dry-dock 5 feet or more above a solid surface (e.g., camel or barge), fall prevention/protection methods must be in place.

a. Personal floatation devices (PFDs) must be worn when working with 6 feet of deck edges of vessels afloat and other surfaces above water (e.g., piers, camels, barges, floating cranes, etc.) where guardrails or an equivalent form of fall protection is not provided to prevent a fall into the water.

3. Aerial Work Platforms (AWP).

a. When working in an AWP, a Fall Restraint System is the safest and preferred method of protecting the occupants. If an adjustable energy absorbing lanyard cannot be configured in a Fall Restraint System, then an energy absorbing lanyard must be configured for Fall Arrest to prevent contact with the nearest lower obstruction.

(1) Self-retracting devices are prohibited from use while working from an AWP.

b. Personnel must stand on the floor of the platform only, must not sit or climb on railings, and must not use planks, ladders, or other devices positioned on railings.

c. When working inside the platform, personnel must not anchor outside of the platform.

d. When exiting/entering the basket at heights, personnel must maintain 100% fall protection. Entry and exit requirements must be outlined in a Fall Protection Plan in accordance with section 220.5.B.2. (Fall Protection Plans).

e. Personnel must follow procedures outlined in reference A. For procedures not addressed within reference A, personnel must follow Original Equipment Manufacturer (OEM) instructions.

(1) When an AWP provides an anchor point, it must be used in accordance with the OEM instructions or as outlined in this chapter, whichever is more stringent.

WARNING
DO NOT tie off to guardrails!

f. When work will take a platform over water, personal floatation devices must be worn by the occupants in addition to their personal fall protection system in accordance with the following:

(1) Where the distance from the working surface to the water is 25 feet or more, personal fall restraint or arrest is required. Personal floatation devices are not required.

(2) Where the distance from the working surface to the water surface is less than 25 feet and the water depth is less than 10 feet, or hazards (i.e., machinery, barges, camels, or other structures) are present, personal fall restraint or arrest is required. Personal floatation devices are not required.

(3) In situations outside of the above scenarios, occupants may disconnect their personal fall protection from the anchor point when the platform is over water. As the operator is trained in both the capabilities of the AWP and as an Authorized Person of fall protection, they, with their supervisor, are in the best position to make this risk decision. This decision will depend on the unit in use, operator experience, height above water, depth of water, wind conditions, and structures that potentially create a tunneling effect of wind.

g. Additional requirements within Chapter 34 of Reference A (OPNAV M-5100.23, Aerial Work Platform Safety) must be followed as applicable.

4. Crane-Suspended Personnel Platform. The lifting of personnel by crane must only be accomplished in accordance with Reference C (NAVFAC P-307).

5. Fixed Ladders General Requirements.

a. The use of ladders with broken or missing rungs or steps, broken or split side rails, or other faulty or defective construction is prohibited. When ladders with such defects are discovered, they shall be reported to Code 106 immediately.

b. Any ladder with structural or other defects must be immediately tagged “Dangerous: Do Not Use” or other similar language, and removed from service.

c. Inspection of metal ladders must include checking for corrosion of interiors of open end, hollow rungs.

d. Ladder surfaces must be free of puncture and laceration hazards.

e. All personnel must face the ladder when climbing.

- f. All personnel must maintain three points of contact on the ladder while climbing (i.e., one hand, two feet; one foot, two hands).
- g. Personnel must not carry any object or load (e.g., tool bag) that could cause them to lose balance and fall while climbing up or down the ladder.
- h. Regardless of fixed ladder length or height, if a fixed ladder possesses a fall arrest system (e.g., SRD or climbing ladder safety system), then this fall arrest system must be utilized when climbing.
- i. Personnel performing work at any height from a fixed ladder must utilize a fall arrest system in conjunction with a fall protection work positioning system.
- j. Additional fixed ladder requirements pertaining to their design can be found within Reference D (29 CFR 1910.23).

6. Fixed Ladders (non-shipboard).

- a. Fall protection is required on fixed ladders that extend over 20 feet in height. Each fixed ladder over 20 feet must be equipped with a personal fall arrest system or ladder safety system.

(1) The 20 feet is measured from the initial walking-working surface where the ladder begins to the next walking-working surface/platform/landing where the ladder ends. The distance of the ladder extension rails above the next walking-working surface is excluded from this measurement.

- b. For fixed ladders equipped with safety cages that were installed before November 19, 2018 and extend more than 20 feet, cages may remain on the ladder; however, they are not recommended as the mitigation measure for a fall.

NOTE

If a fixed ladder starts at a landing that is greater than 20 feet above a lower level, fall protection must be provided regardless of the length of the ladder. For example; if a ladder section begins at a landing and the climber could fall greater than 20' past the landing, but the length of that ladder is less than 20', fall protection must be provided.

- c. Additional fixed ladder requirements pertaining to fixed ladder fall protection can be found in Reference D (29 CFR 1910.28 and 1910.29).

7. Fixed Ladders (shipboard)

- a. In-hull bulkhead ladders (i.e., half-moons, D-hole cutouts) are not recognized as a safe walking-working surface and must not be used without utilizing a Personal Fall Arrest System.

8. Portable Ladders.

a. Personnel must face the ladder and keep their center of mass within the confines of the side rails when performing work from a portable ladder.

b. Personnel performing work from a portable ladder near guardrails with a fall hazard beyond the base of the ladder must utilize a secondary fall protection system. Tie-off to the portable ladder is prohibited.

(1) A personal fall arrest system must be used when personnel are working from a portable ladder near guardrails, such that personnel could fall over the guardrails.

c. Personnel must set up all straight or extension ladders with a 4:1 ratio and lashed, blocked, or otherwise secured to prevent accidental displacement.

d. Neither the top of a stepladder nor the step below the top of a stepladder may be used as a step or used to stand on while performing work.

e. The top two steps of a portable ladder must not be used.

f. Portable ladders used to gain access to an upper landing surface must extend at least 3 feet above the upper landing surface.

g. Portable ladders with structural or other defects must not be used.

D. Passive Fall Protection Systems and Requirements:

1. Guardrail Systems.

a. The top rail of guardrail systems must be capable of withstanding, without failure, a force of at least 200 pounds applied in a downward or outward direction within 2 inches of the top edge, at any point along the top rail.

b. When the 200-pound test load is applied in a downward or outward direction, the top rail of the guardrail system must not deflect to a height of less than 39 inches above the walking-working surface.

c. Guardrail systems must have a top rail height of 42 inches +/- 3 inches.

NOTE

If the working surface adjacent to the guardrail system is elevated, reducing top rail height to less than 39 inches, additional fall protection measures must be in place.

d. The mid rail must be located approximately half way between the top rail and the platform and be capable of supporting, without failure, 150 pounds applied in any downward or outward direction at any point along the intermediate member.

e. A toe board, when installed, must be at least 3 ½ inches high, and must be capable of supporting, without failure, 50 pounds in any downward or outward direction.

f. Flexible guardrail systems, constructed with chains, synthetic fiber, or wire rope must meet the following requirements:

(1) The rope or chain must be secured to each support and taut at all times. It might be necessary to use a positive tensioning device to reduce the sag to a minimum.

(2) The rope or chain must be free of sharp edges, burrs, or projections.

(3) The maximum deflection of the top rail when a load of 200 pounds is applied in any direction at any point on the top rail must not exceed 3 inches in one direction, which includes the free hanging sag in the rope or chain.

g. Additional Guardrail System requirements can be found within Reference D (29 CFR 1910.29).

h. Additional shipboard-specific guarding requirements:

(1) When working in the vicinity of flush manholes and other small openings of comparable size, such openings must be covered per section 220.4.D.3 (Covers), guarded per section 220.4.D.1 (Guardrail Systems), or guarded to a height of not less than 30 inches (i.e., “U” bar guards).

(2) When working around open hatches or other large openings, the edge of the opening must be guarded per section 220.4.D.1 (Guardrail Systems).

(3) Lifeline and liferail systems must meet the applicable requirements within Reference H (NSTM Chapter 600, Hull Outfitting Equipment).

2. Parapets (e.g. roof edge, Bridge Crane Girders). Parapet height must be a minimum of 39 inches to be considered a compliant Passive Fall Protection System. If the height is less than 39 inches, then additional forms of fall protection are required.

3. Covers

a. Covers are used over floor holes and floor openings (refer to Appendix A for definitions).

b. Covers must be capable of supporting at least twice the foreseen weight (i.e., personnel, equipment, supplies).

- c. Covers must be secured to prevent accidental displacement.
- d. Covers must be clearly marked to warn of the hazard (e.g., “HOLE”, “FALL HAZARD”, etc.)

E. Active Fall Protection Systems and Requirements:

1. General Requirements

a. Active Fall Protection Systems must be utilized per the component’s intended application of use per Reference E. Newly acquired fall protection equipment must meet the latest applicable product standard under Reference E. Fall protection equipment not meeting at least ANSI/ASSE Z359.1-2007 is not authorized for use.

b. For Authorized Persons outside of the ANSI recognized weight range of 130-310 lbs., extra precautions must be made. Personal fall protection systems, per ANSI, are certified only within the capacity range of 130 to 310 lbs. The concern for Authorized Persons with a body weight of more than 310 pounds is the potential to fully deploy and bottom-out an energy absorber, exposing the Authorized Person to arrest forces greater than 1,800 pounds prescribed by the Occupational Safety and Health Administration (OSHA), as well as ANSI. The concern for Authorized Persons with a body weight less than 130 pounds is the high arrest force (G force) that they may experience.

Due to the above stated concerns, Authorized Persons outside of the ANSI recognized weight range of 130 – 310 pounds are permitted to use personal fall protection equipment as prescribed by a Qualified Person for Fall Protection as a Fall Restraint System only. Fall Arrest Systems are prohibited for use with Authorized Persons outside of this weight range.

A Qualified Person must develop a Fall Protection Plan meeting the requirements of section 220.5.B.2 (Fall Protection Plans) for Authorized Persons outside of the ANSI recognized weight range.

2. Anchorages

a. All anchorages must be independent of any anchorages used to support or suspend platforms.

b. Certified anchorages

(1) Certified anchorages must have labels (see example Appendix C) and/or documentation stating the following as a minimum:

(a) Anchorage Strength

- (b) Number of personnel allowed to connect
 - (c) Qualified Person for Fall Protection certification signature, date, and contact information
 - (d) Inspection Due Date
 - (e) Permitted uses (e.g., fall arrest, fall restraint, work positioning, etc.)
- (2) Certified Anchorages must:
- (a) Be designed, installed, and used as part of a complete personal fall protection system.
 - (b) Maintain a safety factor of 2:1.
 - (c) Be designed and used under the direction of a Qualified Person for Fall Protection.
- (3) Certified Anchorage design:
- (a) A Professional Engineer (PE) that does not maintain the qualification of a Qualified Person (QP) for Fall Protection may design the anchor point; however, a QP must evaluate the fall protection system in its entirety.
 - (b) A PE may do both given they maintain the qualification of a QP.
- (4) Prior to use, Horizontal Lifeline anchorages, to include anchorage for temporary Horizontal Lifelines, must be designed by a registered Professional Engineer with experience in designing horizontal lifeline systems; or designed by a Qualified Person who has appropriate training and experience.
- c. Non-certified anchorages
- (1) Non-certified anchorages must support the required weight set forth by the designed/directed use of the system.
 - (2) Non-certified anchorages must be selected by a Competent Person.
 - (3) Non-certified anchorage strengths must meet the following strengths per Authorized Person attached:
 - (a) 5,000 pounds for fall arrest,
 - (b) 3,000 pounds for work positioning and rescue, and

(c) 1,000 pounds for fall restraint.

d. In certain situations, an anchorage point cannot be installed while maintaining 100% fall protection. An Authorized Person must establish an anchorage point first for themselves, and then all personnel that will follow. This method, known as “first man-up”, is a last resort and places the Authorized Person at risk when installing an anchorage. This method is an Administrative Control and must be discussed in detail within the Fall Protection Plan signed by the Fall Protection Program Manager.

3. Fall Restraint System.

a. Anchorages must be in accordance with section 220.4.E.2 (Anchorages).

b. The Fall Restraint System must not allow the worker’s center of mass to reach beyond the edge of the walking/working surfaces.

c. Fall Restraint Systems must be used only on horizontal or low sloped surfaces (less than 4:12 pitch).

4. Personal Fall Arrest Systems.

a. Anchorages used for Personal Fall Arrest Systems must be in accordance with section 220.4.E.2 (Anchorages).

b. Connecting means (e.g., Lanyards, Self-Retracting Devices, HLLs, etc.) used in Fall Arrest Systems must have energy absorbing capabilities.

c. Daisy-chaining connecting means is prohibited.

d. Newly acquired fall protection equipment must meet the latest applicable product standard under Reference E. Currently owned fall protection equipment not meeting at least ANSI/ASSE Z359.1-2007 is not authorized for use.

e. Systems must be configured so that the impact forces do not exceed 1,800 pounds.

f. Systems must be configured such that the Authorized Person does not contact a lower level or any obstructions in the event of a fall.

5. Personal Fall Arrest System Clearance Requirements.

a. A Competent Person, Qualified Person, or Fall Protection Program Manager must determine that the required clearance of the Personal Fall Arrest System utilized is less than the available clearance to the nearest obstruction within a Fall Protection Plan meeting the requirements of section 220.5.B.2. (Fall Protection Plans).

b. A personal fall arrest system must be configured such that the employee will not free-fall more than six feet unless the proper energy absorber is used and the system limits the Maximum Arresting Force (MAF) to 1,800 pounds.

c. Swing fall must be considered and mitigated to the greatest extent possible. Swing-drop distance should not exceed 4'. If contact with obstructions could occur during swing fall/swing-drop, then a different mitigation method must be determined.

6. Work Positioning Systems.

a. Anchorages must be in accordance with section 220.4.E.2 (Anchorages).

b. Work Positioning Systems must be configured so that personnel cannot fall more than 2 feet and must be used with a secondary fall arrest system.

c. Secondary Fall Arrest System anchorages must be independent of the anchorages for the work positioning system and must meet the requirements of section 220.4.E.2 (Anchorages).

7. Vertical Lifelines (Rope Grabs).

a. Anchorages must be in accordance with section 220.4.E.2 (Anchorages).

b. Independent vertical lifelines and anchorages are required for each user.

c. Users must ensure that the fall arrestor is compatible with the rope being used.

d. Energy Absorbing Lanyards must not exceed 3 feet in length.

e. Once the work area has been reached fall arrestors must be parked as high above the worker as possible to decrease free fall.

8. Horizontal Lifelines.

a. The concern with horizontal lifelines is the force that can be generated along the length of the lifeline and, ultimately, on end anchorages. Horizontal Lifeline anchorages must be in accordance with section 220.4.E.2.b.(4) (Anchorages – Horizontal Lifelines). Clearance requirements must be provided by the Qualified Person who designed the system. The Qualified Person and/or registered Professional Engineer must follow the applicable design requirements stated within Reference E (ANSI Z359.6).

b. Temporary Horizontal Lifelines must:

(1) Be designed by a Qualified Person or pre-engineered by the OEM.

(2) Adhere to all requirements for installation and use provided by the Qualified Person or OEM.

(3) Be installed, certified, and used under the direction/supervision of a Qualified Person or OEM.

c. Permanent Horizontal Lifelines must:

(1) Be installed, certified, and used under the direction/supervision of a Qualified Person.

(2) Be designed by a Qualified Person that supplies a user guide to define the use, inspection frequency, repair, maintenance details, fall clearances, and fall protection equipment.

(3) Have certification tags installed at the location of the horizontal lifeline.

9. Self-Retracting Devices and Lifelines (SRDs/SRLs).

a. SRDs, excluding those designed with Leading Edge capabilities (SRD-LE), must be used for vertical applications only.

i. When designing fall arrest systems that utilize SRD-LEs, the following hazards must be mitigated:

(1) Obstructions during swing fall

(2) Obstructions during free fall

(3) Friction between the SRD-LE line and the leading edge

b. Additionally, the applicable requirements within sections 220.4.E.4 (Personal Fall Arrest Systems) and 220.4.E.5. (Personal Fall Arrest System Clearance Requirements) must be met when utilizing SRDs.

10. Climbing Ladder Fall Arrest Systems (Ladder Safety Systems).

a. Fall protection systems must be installed on fixed ladders greater than 20 feet in length or installed when ladders are mounted at elevated locations exceeding 20 feet in height.

b. The frontal D-ring of the harness must be connected directly to the traveling sleeve by way of the provided connector on the traveling sleeve or by use of a carabiner.

c. The length between the traveling sleeve and the frontal D-ring connection must not exceed 9 inches.

- d. Free fall must not exceed 2 feet when connecting to the frontal or sternal D-ring.
- e. The system must be positioned to permit personnel to ascend/descend the ladder with minimal manipulation of the traveling sleeve.
- f. Mounting of rigid carriers must be in accordance with OEM recommendations. Mountings for rigid carriers are attached at each end of the carrier, with intermediate mountings spaced, as necessary, along the entire length of the carrier so the system has the strength to stop employee falls.
- g. Mounting of flexible carriers and cable guides must be installed in accordance with OEM instructions. Mountings for flexible carriers are attached at each end of the carrier and cable guides for flexible carriers are installed at least 25 feet apart but not more than 40 feet apart along the entire length of the carrier.
- h. Ladder safety system components must not mix with ladder safety system components from a different manufacturer. Ladder safety sleeves are specifically designed for the ladder safety line/rail/cable that accompanies it.
- i. Ladder safety systems must extend three feet above the accessible walking-working surface.
- j. 100% fall protection must be provided when transitioning while elevated.

11. Connecting Components (e.g., snap hooks, carabiners, etc.)

- a. Component connections must be compatible. It is possible for two components to be compatible with each other when properly configured and used, but to be incompatible when configured and/or used in a different manner (e.g., large snaphooks within AWP's used for fall restraint are compatible, but not compatible when used for fall arrest).
- b. Connections made within any part of a personal fall protection system must be compatible to prevent forced rollout, gate loading, and transverse loading.

F. Accessing Unguarded Rooftops:

1. Prior to accessing a rooftop, permission must be granted by the shipyard's OSHE Office and the Building Manager.
2. Rooftop accesses must be secured and/or appropriately posted to prevent unauthorized access, stating that a Fall Hazard Survey and Protection Plan is required beyond the points of entry.

3. A Fall Hazard Survey and Protection Plan must be conducted for all rooftops where access is needed. Each roof top Fall Hazard Survey and Protection Plan requires two people to complete: (1) a Competent Person to conduct the Fall Hazard Survey and Protection Plan, and (2) a person to serve as the safety observer to ensure personnel stay well away from edges while conducting surveys.

1) Competent Person(s) must not perform work while conducting the survey.

4. Flat roofs or roofs having a pitch of 4:12 or less can be evaluated from the roof surface.

5. Sloped roofs must be evaluated via an AWP when the roof pitch is greater than 4:12 (18.5 degrees), the surface is slippery, or roofing material is fragile.

NOTE

Passive fall protection methods are not acceptable for use on roofs that are greater than 4:12 pitch.

6. Additionally, all roof work fall protection requirements within Reference D (29 CFR 1910.28(b)(13)) must be met.

G. Purchasing, Inspection, Storage, Care, and Maintenance of Fall Protection Equipment:

1. Purchasing.

a. Fall Protection Program Managers must approve the purchase of all fall protection equipment.

b. Fall protection equipment must meet the criteria of Reference E (ANSI Z359).

(1) Electrically rated harnesses must meet the ASTM F887 standard.

(2) When purchasing, the most recent ANSI standard for each component must be specified.

c. All equipment purchased for use in fall protection systems must be properly labeled and serialized for the purpose of tracking required maintenance. New inventory must be entered into an inspection and maintenance program prior to issue.

2. Inspection.

a. Annual Detailed Inspection.

(1) Annual detailed inspections of fall protection equipment must be performed annually from the date the equipment was put into service, unless otherwise specified by the Competent Person, Qualified Person, or Fall Protection Program Manager.

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(2) Annual Detailed Inspections must be accomplished by a Competent Person other than the Authorized Person and documented.

b. Tool Room Re-issue Inspection must be performed:

(1) Prior to reissue to the Authorized Person when the Annual Detailed Inspection is not yet required.

(2) By Competent Persons or Equipment Inspectors who have received knowledge-based training on fall protection equipment inspections in accordance with this chapter.

c. Authorized Person's Inspection.

(1) The Authorized Person must perform a pre-use inspection before accepting or using any fall protection equipment.

(2) The Authorized Person must inspect their issued equipment prior to returning the equipment to the tool room.

d. Fall protection equipment must be inspected for:

(1) Absence or illegibility of markings (e.g., ANSI standards, gate strengths, etc...).

(2) Absence of any elements affecting the equipment form, fit, or function.

(3) Evidence of defects in or damage to hardware elements, including cracks, sharp edges, deformation, corrosion, chemical damage, excessive heating, alteration, and excessive wear.

(4) The inspection must include the snap hooks, carabiners, hardware, cable, chain, rope, webbing, stitch patterns, full body harness, and impact indicators for any defects or unsatisfactory conditions. See Appendix D (Example Inspection Attributes record) for assistance during inspection.

e. Equipment that does not meet the inspection requirements must be removed from service, destroyed, provided to the Fall Protection Program Manager for disposition, or undergo appropriate service prior to reissue/use.

(1) Equipment that has been impacted must be provided to the FPPM for disposition.

f. Inspection records must be maintained for the life of the equipment on all equipment available for distribution.

g. Specialized Fall Protection Equipment.

(1) Some equipment might require the Qualified Person or OEM to perform recertification on specialized equipment (i.e., self-retracting lifelines, automatic descent control devices, ladder climbing devices, and horizontal lifelines). Follow all OEM instructions for recertification. The Fall Protection Program Manager must track recertification.

(2) Unique instructions provided by the OEM, which might arise due to conditions of use, must be retained as a Reference for the life of the equipment, and must be made available to the user.

3. Storage. Store fall protection equipment in a cool, dry, safe environment to prevent damage from environmental factors such as heat, light, excessive moisture, oil, chemicals and their vapors, or other degrading elements.

4. Care and Maintenance.

a. When laundering is required, it must be performed in accordance with the OEM instructions.

b. During annual inspections, all fall protection equipment removed from service or requiring maintenance must be identified by the inspector and notated on an inspection attributes record remark sheet and either provided to the Fall Protection Program Manager for disposition or destroyed. (See Appendix D, Example Inspection Attributes Record).

220.5 RESPONSIBILITIES:

All Shipyards, which have personnel exposed to fall hazards, are required to establish a comprehensively managed fall protection program. The fall protection program must be in writing and approved by Code 106. Per Reference A, a comprehensive managed Fall Protection Program includes the following:

- (1) Command, unit, or activity Policy
- (2) Duties and Responsibilities
- (3) Workplace Surveys and Assessment of Fall Hazards
- (4) Fall Hazard Prevention and Control, including the Preparation of Fall Protection and Prevention Plans.
- (5) Training
- (6) Inspection, Storage, Care, and Maintenance of Fall Protection Equipment
- (7) Rescue Procedures
- (8) Fall mishap reporting
- (9) Audits and Evaluation

A. Subject Matter and Program Requirements:

1. Norfolk Naval Shipyard is the Cognizant Lead Yard (CLY) for this chapter. The CLY must:

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a. Initiate chapter revisions when changes or modifications to Federal, DOD, Navy, or NAVSEA requirements (whether in actual content or interpretation of same) are received which would affect the content of the OSHE Control Manual (OSHECM) chapter.

b. Specify NAVSEA or other Federal technical manuals, instructions, etc., with which revisions are in compliance.

c. Maintain the OSHECM chapter under a three-year revision cycle, as follows:

(1) On an ongoing basis, compile a list of changes that might be required to the chapter.

(2) Each calendar year, prepare a memorandum for CLY OSHE Director review recommending one of three courses of action:

(a) No changes are required; continue to the following year's annual review.

(b) Minor changes are required; determine if the changes are significant enough to proceed with revision or capture the recommended changes and save them for the next revision.

(c) Major changes are required; revise the chapter in accordance with the OSHECM Process. (Refer to OSHE Control Manual, Chapter 101 (Administrative), section 101.4.C.)

(3) At the end of each three-year revision cycle, certify that the chapter is still acceptable as shipyard policy. If the CLY determines the chapter to be acceptable (i.e., relevant), continue to the following three-year revision cycle. If the CLY determines the chapter not to be acceptable, make a recommendation to SEA 04R to cancel the chapter, including the justification for doing so. If the chapter is cancelled, utilize reference A until such revision is complete.

2. All shipyard departments, shops, and codes that perform work within the scope of this document must:

a. Comply with the requirements of this instruction regardless of the site location where personnel are working when fall hazards exist.

b. Delineate and assign responsibilities in the implementation of a managed fall protection program.

c. Ensure that assigned personnel have the necessary skills, knowledge, training, and expertise to manage, administer, and implement the fall protection program.

3. The fall protection program must be organized, as follows:

a. The Fall Protection Program Manager must:

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- (1) Be trained in accordance with section 220.5.C.1 (Fall Protection Program Manager Training).
- (2) Be trained in accordance with section 220.5.C.3. (Competent Person Training).
- (3) Develop, implement, monitor, and evaluate the fall protection program.
- (4) Approve/concur on Fall Hazard Survey and Protection Plans when administrative controls are used.
- (5) Have knowledge of current regulations and standards.
- (6) Provide guidance to Authorized Persons, Competent Persons, Qualified Persons, employees, purchasing, human resources, and engineering on matters pertaining to the fall protection program.
- (7) Ensure that personnel exposed to fall hazards and other personnel involved in the program receive effective/adequate training as outlined in Reference A.
- (8) Participate in all investigations involving mishaps related to falls from heights by reviewing incident reports, taking corrective action to eliminate causes, reporting to management, and maintaining an incident reporting system.
- (9) Conduct periodic program evaluations and ensure improvements are implemented by appropriate organizations.
- (10) Conduct periodic surveillances to validate compliance with this chapter.
- (11) Audit and evaluate the fall protection program annually to validate compliance with this chapter.
- (12) Have access to the QPS/OSHE DR reports quarterly for the purpose of trend analysis.
- (13) Maintain records of the Fall Hazard Survey and Protection Plans, and Rescue Plans submitted by Competent and Qualified Persons.
- (14) Track recertification of specialized fall protection equipment.

b. Qualified Persons for Fall Protection must:

- (1) Serve as an engineer with a recognized engineering degree, or possess a professional certificate and extensive knowledge, training, and experience in fall protection. The Qualified

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Person for Fall Protection must be trained in accordance with section 220.5.C.2 (Qualified Person Training).

(2) Maintain the qualifications of a Competent Person in accordance with section 220.5.C.3 (Competent Person Training).

(3) Support the fall protection program in areas of expertise, including system design, horizontal lifeline design, structural analysis, calculation of arresting forces and fall clearances, testing, anchorage certification, methods of control, equipment selection, administrative controls, and compliance with regulations and standards.

(4) Submit their Fall Hazard Survey and Protection Plans and Rescue Plans to the Fall Protection Program Manager for assessment and recordkeeping.

c. Competent Persons for Fall Protection must:

(1) Be trained in accordance with section 220.5.C.3 (Competent Person Training).

(2) Conduct Fall Hazard Surveys and identify all fall hazards prior to Authorized Persons being exposed.

(3) Develop and implement Fall Protection Plans in compliance with this chapter.

(4) Develop Rescue Plans and pre-incident plans as required.

(5) Participate in all investigations of all falls under their purview.

(6) Have knowledge of current regulations and standards.

(7) Evaluate, conduct, approve and maintain all Fall Hazard Survey and Protection Plans under their purview.

(8) Assist Authorized Persons, and Supervisors of Authorized Persons, as requested.

(9) Submit Fall Hazard Survey and Protection Plans and Rescue Plans to the Fall Protection Program Manager for assessment and recordkeeping.

d. Supervisors of Authorized Persons for Fall Protection must:

(1) Be trained in accordance with section 220.5.C.6 (Authorized Person Training) or 220.5.C.3 (Competent Person Training).

(2) Understand the policies and procedures of this chapter, as well as the direction of Competent and Qualified Persons.

(3) Review jobsite prior to work and request Fall Hazard Survey and Protection Plans and Rescue Plans when fall hazards are identified.

(4) Brief personnel on Fall Hazard Survey and Protection Plans and Rescue Plans as required.

NOTE

It is not the intent of this requirement that Supervisors of Authorized Persons are the only personnel that can brief an Authorized Person. Competent Persons, Qualified Persons, or Work Leaders/Zone Managers familiar with the work and trained as an Authorized Person/Competent Person can also perform a brief on the Fall Hazard Survey and Protection Plan, or Rescue Plan.

(5) Verify employees are qualified before assigning work requiring the use of fall protection equipment.

e. Authorized Persons for Fall Protection must:

(1) Be trained in accordance with section 220.5.C.6 (Authorized Person Training).

(2) Understand the policies and procedures of this chapter, as well as direction of Competent and Qualified Persons.

(3) Report unsafe or hazardous conditions to the supervisor.

(4) Properly inspect, use, store, and care for their fall protection equipment.

(5) Be briefed on the Fall Hazard Survey and Protection Plan, and Rescue Plan as required prior to commencing elevated work.

(6) Have any necessary equipment before accessing an area requiring personal fall protection.

f. Fall Protection Equipment Inspectors must:

(1) Be trained in accordance with section 220.5.C.3 (Competent Person Training) or section 220.5.C.7 (Fall Protection Equipment Inspector Training).

(2) Inspect fall protection and fall rescue equipment in accordance with section 220.4.G.2.

g. All personnel must:

i. Receive fall hazard awareness training, in accordance with section 220.5.C.8 (Employee general awareness training).

ii. Report any worksites that require a Fall Hazard Survey and Protection Plan to a supervisor before accessing the worksite.

B. Reports, Reporting Requirements, and Records:

1. Fall Hazard Surveys and Fall Protection Plans.

a. A Competent Person, Qualified Person or Fall Protection Program Manager must conduct a Fall Hazard Survey of a work location prior to personnel working at unguarded/unprotected heights. Fall Hazard Surveys must contain the following:

- (1) Work location (building, facility, space, etc.)
- (2) Associated hazards within the location.
- (3) Location of the fall hazard(s).
- (4) Existing mitigation measures (e.g., guardrails, covers).
- (5) Fall Hazard Risk Assessment in accordance with Reference J.
- (6) Proposed mitigation measures.
- (7) Anchorage strengths and locations (as applicable).
- (8) Available clearance(s) at the location.
- (9) Rescue plan (as applicable).

2. Fall Protection Plans.

a. Personnel utilizing personal fall protection equipment, or working at unguarded heights, are required to have a Fall Protection Plan. The Fall Protection Plan is equivalent to a permit to work at heights for Authorized Persons. A Competent Person, Qualified Person, or Fall Protection Program Manager must approve the Fall Protection Plan prior to work being executed. The Fall Protection Plan must be kept at the work site at all times. For ease of use, a Fall Hazard Survey and Protection Plan template is available within Appendix E. Fall Protection Plans must include:

- (1) Description of the fall hazards that will be encountered by Authorized Persons.

(2) Type of Fall Protection/Prevention methods or systems used for every phase of work.

(a) If Administrative Controls are selected, further approval from the Fall Protection Program Manager is required. Furthermore, a reason for the infeasibility of a conventional secondary fall protection system must be given.

(3) Validation of training requirements for Authorized Persons.

(4) Type of fall protection equipment and systems provided to the employees that might be exposed to fall hazards.

(5) Identification of anchorages (certified and/or non-certified, location(s)).

(6) Complete setup procedure(s) for access.

(7) Clearance requirements for free-fall distances and for total fall distances including available and required clearance.

(8) Instructions for assembling, use and dismantling of the system(s), including descriptions of all the components.

(9) Any limitations of the system.

(10) Applicable manufacturers' standards and drawings.

(11) The name of Competent and/or Qualified Person that developed the plan.

(12) Rescue plan and procedures (required for Fall Arrest Systems).

3. Rescue Plans

a. When Authorized Users are utilizing a Fall Arrest System a Rescue Plan must be in place. Rescue Plans are a part of the Fall Hazard Survey and Protection Plan and must be site-specific. In the event of a fall emergency services must be contacted. Rescue Plans must address the following:

(1) Procedure for contacting emergency services. This must take into account:

(a) Exact location of the fallen individual.

(b) Whether a technical rescue is needed. If the individual is suspended or rescue at heights is needed, state that a technical rescue is needed.

(c) Condition of the suspended individual (e.g., conscious vs. unconscious, able to self-rescue, injuries sustained, etc.)

(2) Type(s) of rescue.

(a) Self-Rescue (e.g., Authorized Person is able to safely pull themselves to the nearest structure and rescue themselves, Authorized Person utilizes descent control device).

(b) Assisted Rescue (e.g., stationing an AWP onsite to rescue a suspended individual, providing a portable ladder for the suspended individual, utilization of descent control devices, etc.).

(c) Jurisdictional Public or Government Emergency-Response Agency Rescue.

b. Jurisdictional Public or Government Emergency-Response Agency Rescue requires the development of a pre-incident plan per Reference A. Detailed discussion on pre-incident planning can be found within Reference I.

4. Certain existing documents may be used as a Fall Hazard Survey and/or Fall Protection Plan given they meet the applicable requirements within this section. Examples of existing documents that may serve as a Fall Hazard Survey/Fall Protection Plan include, but are not limited to, Industrial Process Instructions, Technical Work Documents and Lifting & Handling Instructions.

5. Mishaps or deviations from requirements that might result in damage to Navy facilities and equipment, occupational injuries, illnesses, or death must be immediately reported to the appropriate supervisor and OSHE Office. In addition, an OSHE Deficiency Report (DR) must be documented, when required by OSHE Control Manual, Chapter 130 (OSHE Deficiency Reporting System).

6. All falls, regardless of impact loading and/or injury, are required to be reported.

7. Surveillance findings must be entered into the Quality Performance System (QPS) for the purpose of trend analysis and determining the health of the Fall Protection Program.

8. Safety Trouble Reports (TR) must be submitted for any fall from any height greater than 4 feet for general industry work and 5 feet for shipyard repair operations in accordance with Reference G (NAVSEAINST 4700.17).

C. Training:

The following training requirements must be met:

1. Fall Protection Program Manager training:

a. Fall Protection Program Managers must receive training per Reference A.

2. Qualified Person training:

- a. Qualified Persons for Fall Protection must receive training per Reference A.
- b. Qualified Persons must maintain the qualification of a Competent Person.
- c. Qualified Persons must receive refresher training at intervals specified by the training establishment; in order to stay current with fall protection, rescue and technical requirements.

3. Competent Person training:

- a. Competent Persons for Fall Protection must receive training per Reference A.
- b. Competent Persons must receive refresher training every two years; in order to stay current with fall protection, rescue and technical requirements.
- c. Initial Competent Persons training must be a minimum of 40 hours. Refresher Competent Persons Training must be a minimum of 16 hours.

NOTE

In regards to Competent and Qualified Person refresher training; if an employee is unable to receive refresher training within the timeframe of the above stated requirements, the C106 Fall Protection Program Manager may provide an extension due to extenuating circumstances. This extension is at the discretion of the Fall Protection Program Manager and should be determined on a case-by-case basis, but must not exceed 90 days.

4. Competent Person Instructor training:

- a. Competent Person Instructors must maintain the qualification of a Competent Person by attending an Echelon II approved course.
- b. Competent Person Instructors must receive refresher training every two years in order to stay current with fall protection, rescue and technical requirements.

5. Authorized Person Instructor training:

- a. The Authorized Person Instructor must receive training as a Competent Person in accordance with section 220.5.C.3 (Competent Person Training).

6. Authorized Person training:

- a. Authorized Persons must receive training per Reference A.

b. Authorized Person skill-based practical (hands-on) training must be provided by a Competent Person for Fall Protection and include demonstrations by students on how to inspect, anchor, assemble, and use the fall protection equipment as used in locations where they work. Prior to use, personnel must receive training on the specific type of fall protection equipment to be used.

c. Authorized Persons must receive refresher training every two years to stay current with fall protection and rescue educational requirements.

d. Authorized Persons must receive retraining, by a Competent Person, in accordance with Reference A.

7. Fall Protection Equipment Inspector Training:

a. Fall Protection Equipment Inspector training must include:

(1) Specific inspection attributes of all fall protection equipment used at the shipyard per the applicable ANSI standard and OEM manual.

(2) Designated process for documenting inspections.

b. Fall Protection Equipment Inspectors must receive refresher training at a minimum of every two years.

NOTE

An equipment inspector maintaining the qualification of a Fall Protection Competent Person satisfies the training requirements of a Fall Protection Equipment Inspector.

8. Employee general fall hazard awareness training:

a. All personnel that may be exposed to fall hazards must receive general awareness training annually.

b. Employee general fall hazard awareness knowledge-based training must include:

(1) Applicable regulations and standards.

(2) Safe work practices.

(3) Safe access to elevated heights.

(4) Fall hazard recognition.

(5) Personal protective equipment.

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- (6) Walking-working surfaces.
- (7) Scaffolds, ladders, and other working surfaces.
- (8) Reporting a hazard.
- (9) When a Fall Hazard Survey and Protection Plan is required.

Definitions

Activation distance	The distance traveled by a fall arrestor or the amount of line paid out by a self-retracting lanyard (SRL) from the point of onset of a fall to the point where the fall arrestor or Self-Retracting Lifeline (SRL) begins to apply a braking or stopping force.
Administrative controls	Indicate that a worker is exposed to a fall hazard and outline, within a formal fall protection plan, how to perform work tasks as safely as possible without a secondary system (e.g., erecting warning lines, or designated areas, restricting access to certain areas or posting warning signs).
Anchorage	A secure connecting point or a terminating component of a fall protection or rescue system capable of safely supporting the impact forces applied by a fall protection system or anchorage subsystem.
Anchorage connector	A component or subsystem that functions as an interface between the anchorage and fall protection, work positioning, rope access, or rescue system for the purpose of coupling the system to the anchorage.
Anchorage subsystem	A subsystem of a complete active fall protection system to which workers connect their personal equipment (e.g., vertical lifelines, horizontal lifelines, ladder safety systems, rigid rails).
Arrest distance	The total vertical distance required to arrest a fall. This arrest distance includes the deceleration distance and the activation distance.
Authorized Person	A person assigned and trained to use fall protection equipment by the employer to perform duties at locations exposed to fall hazards.
Available clearance	The distance from a Reference point, such as the working platform, to the nearest obstruction which, if struck during a fall, could cause injury.
Carrier	The track of a ladder safety system consisting of a flexible cable or rigid rail, which is secured to the ladder or structure.
Competent Person	An individual through training and knowledge who is capable of identifying, evaluating, and addressing existing and potential fall hazards and who has been designated by the employer to be responsible for the immediate supervision, implementation, and monitoring of the managed fall protection program and who has the authority to take prompt corrective action with regard to such hazards.
Conventional Fall Protection	Any secondary fall protection systems that are not stated within the Administrative Controls definition (i.e., Passive Fall Protection Systems, Fall Restraint Systems, and Fall Arrest Systems).
Cover	A guard for a floor opening/floor hole which is capable of supporting, without failure, at least twice the weight of employees, equipment, and materials that could be imposed onto it at any one time.

Dangerous equipment	Equipment, such as vats, tanks, electrical equipment, machinery, equipment or machinery with protruding parts, or other similar units, that, because of their function or form, may harm an employee who falls into or onto the equipment.
Deceleration distance	The vertical distance between the user's fall arrest attachment at the onset of fall arrest forces during a fall, and after the fall arrest attachment comes to a complete stop.
Dockboards	A portable or fixed device that spans a gap or compensates for a difference in elevation between a loading platform and a transport vehicle. Dockboards include, but are not limited to, bridge plates, dock plates, and dock levelers.
Energy Absorber	A component whose primary function is to dissipate energy and limit deceleration forces which the system imposes on the body during fall arrest.
Engineered anchorage (i.e., Certified anchorage)	An anchorage for fall arrest, positioning, restraint, or rescue systems certified by a Qualified Person, engineer, and/or Professional Engineer as capable of supporting the necessary loads in the event of a fall.
Extension ladder	A non-self-supporting portable ladder adjustable in length. It consists of two or more sections traveling in guides or brackets so arranged as to permit length adjustment. Its size is designated by the sum of the lengths of the sections measured along the side rails.
Failure	A load refusal, breakage, or separation of component parts. A load refusal is the point at which the ultimate strength of a component or object is exceeded. When referring to guardrail systems testing; failure occurs when the system deflects to a point that renders the system out of compliance.
Fall arrest	The action or event of stopping a free fall or the instant where the free fall has been stopped.
Fall arrestor	A device that travels on a lifeline and will automatically engage or lock onto the lifeline in the event of a fall.
Fall hazard	Any condition on a walking-working surface with unprotected sides, edges, or floor openings, from which there is a possibility of falling on the same level, to a lower level, onto dangerous equipment, into a hazardous environment, or onto an impalement hazard.
Fall hazard analysis	The process of Competent or Qualified Person(s) performing onsite fall hazard surveys, as well as developing fall protection plans for individual worksites.
Fall Hazard Survey report	Documentation of a fall hazard survey that contains information about existing or potential fall hazards, risk assessment, fall protection plans (as applicable) and a method or methods for eliminating or controlling those hazards.

Fall Protection Brief	A document or discussion referring to the Fall Hazard Survey/Fall Protection Plan provided to Authorized Persons and personnel exposed to fall hazards.
Fall Protection Plan	Document that includes written procedures for performing a specific work, task, or project, indicating the proper way of using safe Fall Protection systems and equipment, and including any other relevant information.
Fall restraint	The technique of securing an Authorized Person to an anchorage using a lanyard short enough to prevent the person's center of mass from reaching the fall hazard.
First man up	If a fall protection anchorage point has not previously been established, the first person exposed to the fall hazard must establish fall protection for themselves and all Authorized Persons to follow.
Fixed ladder	A ladder permanently attached to a structure, building, or equipment.
Floor hole	An opening measuring less than 12 inches, but more than 2 inches in its least dimension, in any floor, platform, pavement, or yard, through which materials but not persons, may fall, such as a belt hole, pipe opening, or slot opening.
Floor opening	An opening measuring 12 inches or more in its least dimension, in any floor, platform, pavement, or yard through which persons may fall, such as a hatchway, stair or ladder opening, pit, or large manhole.
Free fall	The act of falling before a fall protection system begins to apply forces to arrest the fall.
Guardrail system	A passive system of horizontal rails and vertical posts that prevents a person from reaching a fall hazard.
Handrail	A single bar or pipe supported on brackets from a wall or partition, as on a stairway or ramp, to furnish persons with a handhold in case of tripping.
Harness, full body	A body support designed to contain the torso and distribute the fall arrest forces over at least the upper thighs, pelvis, chest, and shoulders.
Hazard elimination	Changing the task, process, controls, or other means so as to remove the need for an employee to be exposed to a fall hazard.
Hazardous environment	An environmental condition such that, if an employee were to fall in, would cause severe injury and/or severe health complications (e.g., corrosive liquid containers, toxic environments).
Hoist area	Any elevated access opening to a walking-working surface through which equipment or materials are loaded or received.
Horizontal lifeline	A stationary lifeline strung between two anchorage points on a horizontal plane and used as an attachment point for personal fall arrest/restraint systems.
Impalement hazard	Protruding objects that would impale an employee if they were to fall.

Infeasible	Impossible to perform the work using a conventional fall protection system (i.e., passive or active fall protection systems), or technologically impossible to use any one of those systems to provide fall protection.
Isolated location	An area outside of the sight and sound of another person.
Ladder cage	An enclosure that encompasses a fixed ladder by encircling the climbing space of the ladder for the safety of the person who must climb the ladder.
Ladder safety system	A system designed to eliminate or reduce the possibility of falling from a ladder. A ladder safety system usually consists of a carrier, safety sleeve, connectors, and body harness. Cages and wells are not ladder safety systems.
Lanyard	A component consisting of a flexible rope, wire rope, or strap, which typically has a connector at each end, for connecting to the body support and to a fall arrestor, energy absorbing anchorage connector, or anchorage.
Lifelines	Removable safety barriers rigged along deck edges and deck openings and in areas from which all obstructions must be readily removable for action, exercises, or operations.
Liferails	Rigid barrier, either permanently fixed or hinged/portable, installed along deck edges or platforms.
Low Slope surface	A roof or surface that has a slope less than or equal to a ratio of 4:12 (vertical inches to horizontal inches).
Maximum arrest force	The peak force measured by the test instrumentation during arrest of the test weight in the dynamic tests set forth in American National Standards Institute (ANSI). Maximum arresting force must be limited to 1,800 lbs.
Non-engineered anchorage (i.e., non-certified anchorage)	An anchorage, not based on a Qualified Person calculation, that a Competent Person can judge to be capable of supporting the predetermined anchorage forces outlined within this chapter.
One hundred percent tie-off	A requirement that a personal fall arrest system must be connected to an anchorage 100% of the time either with dual lanyards, self-retracting lifelines, or a combination of the two.
Parapet	A low protective wall or barrier at the edge of a balcony, roof, bridge, bridge crane, or the like.
Passive fall protection	Fall protection that does not require the wearing or use of personal fall protection equipment.
Personal Fall Arrest System (PFAS)	An assembly of components and subsystems used to arrest a person in a free fall.

Personal Fall Protection (Equipment)	A system (including all components) an employer uses to provide protection from falling or to safely arrest an employee's fall if one occurs. Examples of personal fall protection include personal fall arrest systems (PFAS), work positioning systems, and fall restraint systems.
Platform	A working space for persons, elevated above the surrounding floor or ground, such as a balcony or platform for the operation of machinery and equipment.
Portable ladders	A ladder that can readily be moved or carried, usually consisting of side rails joined at intervals by steps, rungs, cleats, or rear braces.
Primary fall protection system	The system supporting a person's weight while performing work at height, used in conjunction with a secondary system. (e.g., the walking-working surface a person is on, work positioning systems).
Professional Engineer	A person who holds an engineering registration in the state or other jurisdiction in which they are applying these standards.
Purview	The scope of the influence or concerns of something. A range of experience or thought.
Qualified Person	A person with a recognized degree or professional certificate and/or with extensive knowledge, training, and experience in the fall protection and rescue field who is capable of designing, analyzing, evaluating, and specifying fall protection and rescue systems to the extent required by this program.
Related employment	Any employment performed as an incident to or in conjunction with ship repairing, shipbuilding, or shipbreaking work, including, but not restricted to, inspection, testing, and employment as a watchman.
Rescue plan	A written process that describes in a general manner how rescue is to be approached under the specified parameters, such as location or circumstances.
Rollout	A form of failure by which a snap hook or carabiner unintentionally disengages from another connector or object to which it is coupled.
Rope grab	A deceleration device that travels on a lifeline and automatically, by friction, engages the lifeline and locks so as to arrest the fall of an employee. A rope grab usually employs the principle of inertial locking, cam/lever locking, or both.

Secondary fall protection system	One or more means of fall protection configured as a supplement or backup to a primary system for the purpose of providing protection from a potential fall. (e.g., Self-Retracting Lifeline (SRL), ladder safety systems, or rope grab).
Self-retracting device (SRD)	A device containing a drum wound line that automatically locks at the onset of a fall to arrest the user, but that automatically pays out from and retracts onto the drum during normal movement of the person to whom the line is attached. After onset of a fall, the device automatically locks the drum and arrests the fall. SRDs include Self-Retracting Lanyards (SRL), Self-Retracting Lanyards with integral rescue capability (SRL-R), self-retracting lanyards with leading edge capabilities (SRL-LE), and, hybrid combinations of these. NOTE: Self-retracting “lifelines and lanyards” terminology is used interchangeably.
Must	A mandatory requirement.
Shipyard operations	Ship repair, shipbuilding, or shipbreaking, or all related employment.
Ship repair	Any repair of a vessel including, but not restricted to, alterations, conversions, installations, cleaning, painting, and maintenance work.
Should/May	A recommendation.
Stairwell	A series of steps and landings having three or more risers leading from one level or floor to another, or leading to platforms, pits, boiler rooms, crossovers, or around machinery, tanks, and other equipment that are used more or less continuously or routinely by employees, or only occasionally by specific individuals.
Step ladder	A self-supporting portable ladder, nonadjustable in length, having flat steps and a hinged back. Its size is designated by the overall length of the ladder measured along the front edge of the side rails.
Swing-drop distance	The vertical drop in height experienced by the worker using a fall arrest system from the onset of the swinging motion to the point where the user can initially make contact with a structure.
Swing fall	A pendulum like motion that occurs during and/or after a vertical fall. A swing fall results when an Authorized Person begins a fall from a position that is located horizontally away from a fixed anchorage.
Total fall distance	The total vertical distance a person falls, measured from the onset of a fall to the point where the person comes to rest after the fall is stopped.

Transiting	The act or fact of passing across or through; passage from one place to another.
Transverse (sideways) loading	For connecting components, transverse loading occurs when the component is loaded perpendicular to its length.
Traveling sleeve	Part of a ladder safety system consisting of the moving component with locking mechanism that travels on the carrier and makes the connection between the carrier and the harness.
Unguarded edge	Any edge in which a fall hazard exists and adequate fall prevention/protection methods are not in place.
Vertical lifeline	A length of rope with a manufactured termination at the top end and some means to tension the line, such as a small weight at the bottom end. VLLs must only be used with a properly matched fall arrestor, as specified by the manufacturer.
Vertical/wall opening	An opening at least 30 inches high and 18 inches wide, in any wall or partition, through which persons may fall; such as a yard-arm doorway or chute opening.
Work positioning system	A full body harness system configured to allow an Authorized Person to be supported on an elevated vertical or inclined surface, such as a wall, and work with both hands free from body support.

**Excerpt from OSHA Directive Number CPL 02-00-162
(Shipyard Employment “Tool Bag” Directive)**

In this appendix, the applicability of each 29 CFR Part 1910 standard as it pertains to shipyard employment on vessels (including vessel sections) and shore-side operations (land-side operations, excluding vessels and vessel sections) is indicated with a “Yes”, “Yes*” (partial coverage), or “No.” Additionally, "Remarks/Comments" are provided to further clarify the applicability of the 29 CFR Part 1910 standards, including a Reference to any 29 CFR Part 1915 standards that generally take precedence pursuant to 29 CFR 1910.5, Applicability of Standards.

29 CFR Part 1910, Subpart D: Walking-Working Surfaces

Part 1910 Standard	Applicability: Vessel Shore		Remarks/Comments
<u>.28</u>	Yes*	Yes	1910.28 applies to fall and falling object hazards on shore. 1915.71 applies to fall and falling object hazards on scaffolds on shore and 1910.28 applies to fall and falling object hazards on scaffolds that 1915.71 does not address. 1915.71 applies to fall and falling object hazards related to scaffolds on vessels and 1915.73 applies to guarding deck openings and edges. 1910.28 applies to fall and falling object hazards on vessels that 1915.71 and 1915.73 do not address. NOTE: In 1915.73(d), where the nature of the work or the physical conditions prohibit the use or installation of guardrails, employers must provide fall protection in accordance with 1915.159 or 1915.160. In addition, employers must provide personal flotation devices in compliance with 1915.158. NOTE: For shipbreaking operations, 1915.73 does not apply to the guarding of deck openings and edges. Employee exposure to guarding hazards for shipbreaking operations must be addressed by applying 1915.152(b) Hazard assessment and equipment. Fall protection in accordance with 1915.158, 1915.159 and 1915.160, as appropriate, also may be required.
<u>.29</u>	Yes*	Yes	1910.29 applies on shore. 1915.71 applies to scaffolds on vessels, and 1910.29 applies to criteria for fall and falling object protection for scaffolds that 1915.71 does not address. 1915.71, 1915.73, 1915.74, 1915.75, 1915.77, 1915.159 and 1915.160 apply on vessels. 1910.28 applies on vessels to criteria for fall and falling object protection that those sections of 1915 do not address.
<u>.140</u>	No	No	1915.151(a), 1915.159, and 1915.160 applies on vessels and on shore, preempting this personal fall protection system standard in its entirety.

Example Engineered Anchorage Label

Engineered Anchorage		
Description: _____ _____		
Connection Means: _____		
Anchorage Strength: _____		
Number of Persons: _____		
CP Insp. Due Date: _____		
QP Recert. Due Date: _____		
QP Cert. Sig.: _____		
_____	_____	_____
Name	Badge	Date

If unsure how to use anchorage Contact:		
Qualified Person (Print): _____		
Phone Number: _____		

Example Inspection Attributes Record

INSPECTION RECORD						
EQUIPMENT TYPE:		EQUIPMENT SERIAL NUMBER:	DATE INSPECTED:			
ITEM NO.	ITEM	INSPECTION ATTRIBUTES	SAT	UNSAT	N/A	INITIAL
1.	Labels	Must be present and fully legible.				
2.	Rope/Webbing/ Stitching:	Must be free of frayed cut or broken fibers; check for tears, abrasions, mold, burns, or discoloration; must be free of knots, excessive soiling, heavy paint build up, and rust staining; check for chemical or heat damage indicated by brown, discolored, or brittle areas; check for ultraviolet damage indicated by discoloration and the presence of splinters or slivers on the webbing surface; ensure no pulled or cut stitching.				
3.	Hardware (Snap hook, buckles, D-rings, lanyard keepers, chain, pivotal link, ferrules, rivets, etc.)	Must not be damaged, broken, distorted, and must be free of sharp edges, burrs, cracks, worn parts, or corrosion. PVC coated hardware must be free of cuts, rips, tears, holes, etc. in the coating to ensure non-conductivity. Ensure buckles work smoothly. If present, inspect the quick connect buckles by ensuring that the release tabs work freely and that a click is heard when the buckle engages. Inspect parachute buckle spring.				
4.	Impact Indicators	Stitched Impact Indicator: is a section of webbing that is lapped back on itself and secured with a specific stitch pattern holding the lap. The stitch pattern is designed to release when the harness arrests a fall or has been subjected to an equivalent force.				
		Dorsal D-ring Indicator: consists of a D-ring, which is fitted into a plastic housing. This is designed so that the D-ring will be released from the housing exposing a red impact indication label when the harness arrests a fall or has been subjected to an equivalent force.				
		Snap Hook Indicator: The indicator is located on the snap hook or anchorage connection point and deploys either by shearing pin or by bending tabs. Ensure indicator is not deployed.				
5.	Wire Cable	Entire length of wire rope must be inspected for broken strands by flexing every few inches to expose breaks; must be free of corrosion.				
6.	Energy Absorbing Component	Must have no evidence of elongations; cover must be secure with no tears or damage.				
7.	SRL Housing	Must be free of distortion, cracks, or other damage; lifeline must fully extend and retract smoothly with no hesitation or slack in the line; device must lock without slipping when lifeline is pulled sharply.				
8.	General Condition	Ensure there are no loose screws, bent or damaged parts; all equipment is in working condition; clean.				

INSPECTOR SIGNATURE: _____ DATE _____

All deficient conditions must be marked UNSAT (U) on this sheet and identified/described on the inspection attributes remarks sheet.

Example Inspection Attributes Remarks Sheet

EQUIPMENT TYPE		EQUIPMENT SERIAL NUMBER	DATE INSPECTED	
ITEM NO.	DEFICIENCIES IDENTIFIED	CORRECTED SIGNATURE/ DATE	DISPOSED SIGNATURE/ DATE	

FALL HAZARD SURVEY AND PROTECTION PLAN: This document is a standard Fall Hazard Survey and Protection Plan outline for the shipyards. When completed, this plan will summarize the expectations for the Authorized Persons and Supervisors using this plan. All personnel supervising and using fall protection equipment must demonstrate knowledge and understanding of the equipment and requirements of this plan.		
Project/Ship/Building – Detailed Location	Origination Date	Review Date – annually from Origination Date
Specific Work Location/Area		Department/Code/Shop
Qualified Person/Competent Person (Print and Signature):		
Survey Performed By:		
Fall Protection Program Manager (Signature) – Only required when Administrative Controls are in use:		
Fall Hazard Survey		
Fall Hazard/Proximity to Fall Hazard:		
Existing Fall Hazard Mitigation Measures:		
Description of Work Task(s) and Associated Hazards:		
Chance of Same-Level Slips, Trips, or Falls? Yes or No	Location of Slip/Trip Hazards:	
Available Clearance to Nearest Obstruction/Lower Level from Walking-Working Surface:		
Probability of falling:	Severity of fall:	Pre-Mitigation Risk Assessment Code:
Obstructions in Potential Fall Path:		
Access/Egress to Fall-Hazard Area:		
Condition of Work Surfaces (strength, structural integrity, etc.):		
Anchorage location(s), strength(s), certified/non-certified, and use(s):		
Fall Protection Equipment to be used:		

Fall Protection Survey/Plan Continued		
Fall Protection Plan Specifics		
Method	Yes, No, N/A	Describe Exact Location, Equipment set-up, Required Clearances (as applicable), System Limitations (N/A where not applicable, provide sketches if necessary)
Hazard Elimination		
Passive Fall Protection Systems		
Fall-Restraint System		
Fall-Arrest System		Considerations: Free Fall, Required Clearance, Swing Fall, System Limitations
Administrative Controls		
Rescue Plan (required for Fall Arrest)		Considerations: Technical Rescue, Assisted Rescue, Conscious/Unconscious, Specific Location, Emergency Services Phone Number
Training Requirements verified (as applicable):		
Additional information:		
Post-Mitigation Risk Assessment:		
Probability of fall:	Severity of fall:	Risk Assessment Code:
Fall Protection Plan Must Be Approved by a Competent Person or Qualified Person Prior to starting work		